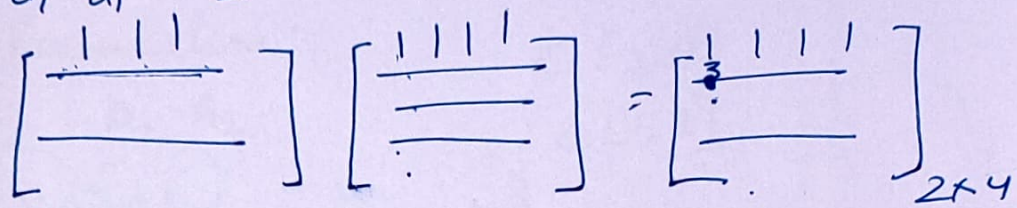
① **Matrix Chain Multiplication**

$\frac{2(n-1)C_{n-1}}{n}$

A_1 A_2

(2×3) (3×4)

d_0 d_1 d_1 d_2



2×4

Multiplication Number

$C[1\,2] = 2\times3\times4 = d_0 d_1 d_2$

2×3 3×4 4×5

A_1 A_2 A_3

d_0 d_1 d_1 d_2 d_2 d_3

$(A_1A_2)A_3$

$(d_0d_1d_2)_{d_0d_2}$ $(A_3)_{d_2d_3}$

$d_0d_1d_2 + d_0d_2d_3$

$2\times3\times4 + 2\times4\times5$

64

$A_1(A_2A_3)$ [Associative]

$(A_1)_{d_0d_1}$ $(d_1d_2d_3)_{d_1d_3}$

$d_1d_2d_3 + d_0d_1d_3$

$3\times4\times5 + 2\times3\times5$

90

$C[1\,3] = \min\{d_0d_1d_2 + d_0d_2d_3,\ d_1d_2d_3 + d_0d_1d_3\}$

(2)

$$c[i,j] = \min_{i \le k < j} \{ c[i,k] + c[k+1,j] + d_{i-1} \times d_k \times d_j \}$$

$A_1\ A_2\ A_3\ A_4\ A_5$

$c[1,5]$
$c[3,4]$
$c[1,1]$

Recursive Formulation:

$A_1\ A_2$

$i\ 1\ \ 2 = j$

$k = 1$

$c[1,1] + c[2,2] + d_0 \times d_1 \times d_2$

$0 + 0 + d_0 d_1 d_2$

$A_1\ A_2\ A_3$

$i = 1,\ j = 3$

$k = 1$

$c[1,3] = \{ c[1,1] + c[2,3] + d_0 d_1 d_3 \}$

$k = 2$

$= c[1,2] + c[3,3] + d_0 d_2 d_3$

$c[1,2] = d_0 d_1 d_2$

$c[2,3]$ $\quad k=2$ $= c[2,2] + c[3,3] + d_1 d_2 d_3$

$= d_1 d_2 d_3$

$c[1,3] = d_1 d_2 d_3 + d_0 d_1 d_3$

$= d_0 d_1 d_2 + d_0 d_2 d_3$

③

$A_1 \times A_2 \times A_3 \times A_4$

d_0	d_1	d_2	d_3	d_4
3	2	2	4	4	2	2	5

$$C[1,4] = \min_{k=1,2,3} \begin{cases} k=1 \rightarrow C[1,1] + C[2,4] + d_0 d_1 d_4 \\ k=2 \rightarrow C[1,2] + C[3,4] + d_0 d_2 d_4 \\ k=3 \rightarrow C[1,3] + C[4,4] + d_0 d_3 d_4 \end{cases}$$

$$C[i,j] = \begin{cases} 0 & i = j \\ \min_{i \le k < j} \{ C[i,k] + C[k+1,j] + d_{i-1} \times d_k \times d_j \} & i \neq j \\ & 1 \le k < j \end{cases}$$

(4)

C	1	2	3	4
1	0	24	28	58
2		0	16	36
3			0	40
4				0

$C[1,2] = d_0 d_1 d_2$
$k=1 \quad = 3 \times 2 \times 4$
$= 24$

S	1	2	3	4
1		1	1	3
2			2	3
3				3
4				

$C[2,3] = d_1 d_2 d_3$
$k=2 \quad = 2 \times 4 \times 2$
$= 16$

$C[3,4] = d_2 d_3 d_4$
$k=3 \quad = 4 \times 2 \times 5$
$= 40$

$(A_1)(A_2\ A_3)(A_4)$

$((A_1)(A_2\ A_3))(A_4)$

$$C[1,3] = \min \begin{cases} k=1: & C[1,1] + C[2,3] + d_0 d_1 d_3 \\ & 0 + 16 + 3 \times 2 \times 2 \\ & 16 + 12 = 28 \\ k=2: & C[1,2] + C[3,3] + d_0 d_2 d_3 \\ & 24 + 0 + 3 \times 4 \times 2 \\ & 24 + 24 = 48 \end{cases}$$

⑤

Algo Matrix Multiplication

C[n x n] S[1,--n-1; 2 --n]

```
for i = 1 to n
    m[i, i] = 0
for l = 2 to n
    for i = 1 to n-l+1
        j = i + l - 1
        C[i, j] = ∞
        for K = i to j-1
            q = C[i, K] + m[K+1, j] + Pi-1 PK Pj
            if q < C[i, j]
                C[i, j] = q
                s[i, j] = K
```

$O(n^3)$

Return m and s